

Facial Emotion Recognition and Behavior Modeling Using Deep Learning

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ABSTRACT

In this study, the author proposes a facial expression recognition and behavior modeling system that is aimed at determining the involvement of students during online learning sessions. With the spread of the digital platform of learning, such as Zoom, being more widespread, real-time attention monitoring of students is one of the most important issues in the educational technology field. The purpose of this study is to find out the engagement level which is divided into three categories namely not engaged, partially engaged, or fully engaged class using face instant photos that have been taken during virtual lectures. In order to create engaging, the detection scheme with the help of frames and video-level deep learning models was proposed using the DAiSEE dataset. It was re-labeled, and only the classes concerning engagement came in focus, which allowed balancing the data used and simplify the learning procedure. The frame-level models (i.e., ResNet50 and EfficientNet-B0) were trained to derive the spatial features and video-level models (i.e., X3D-S, TSM with resnet, Swin Transformer and ViViT) were trained to capture the temporal dynamics of student behavior. The existence of low-resolution video frames is one of the major dilemmas in the analysis of engagement in the online context because it may blur fine details of the facial signal. To counter this, on the test phase the Real-ESRGAN super-resolution model was applied to bring more clarity to the faces. The experimental data demonstrate that this method will be an effective way to increase accuracy of engagement classification, especially when it is necessary to identify fully engaged students. The research can be added to the areas of affective computing and intelligent tutoring systems with an ability to solve the engagement analysis, taking into consideration a scalable data-driven model. It also provides a comparative overview of some of the recent deep learning architectures, indicates the impact of quality visual input on the recognition rates, and defines the foundation of the real-time and automated engagement monitoring systems in the learning context. The proposed system may be implemented in the field of remote learning with solid prospects, as it has overcome some of the most common issues (dataset imbalance and video quality limit), difficult to overcome in practice.

Keywords: Student facial expression system, Expression feedback system, behavior modeling, deep learning, Online facial Expression System.

Chapter 1

Introduction

1.1 Background

Affective computing is a developing interdisciplinary science that combines artificial intelligence, psychology, and human-computer interaction to make machines aware of human feelings and reactions. To develop adaptive learning environments in an educational environment, it is important to detect emotional and cognitive states, e.g. engagement by students. Even though the monitoring in a traditional classroom setting requires the direct interaction between teachers and students, this reduction is lost in online or hybrid learning environments. The recent outbreak of online education, particularly after the COVID-19 pandemic, has made the appearance of issues more urgent that have to be solved, involving the assistance of automated systems that can automatically identify when learners are eager to learn new information and give feedback, and provide help lines immediately. Facial expression analysis is a non-invasive and information-dense method of recognizing emotions, since humans use faces to convey a lot of information about their feelings. Based on recent advancements in deep learning, specifically in the field of Convolutional neural networks (CNN) and transformer-based models, the existing sequence of methods has proved highly successful in speeding up the emotion recognition systems. Nevertheless, the processes of using these techniques in real-life online education situations are still difficult due to the noise in the data, blurry traces of webcam video recording, and undefined outputs in the labeling of the dataset. A great urgency is felt today to come up with viable and expanded

face engagement determination systems that not only aggregate imagery image perception amplification methods but also amplify time coding measures.

1.2 Problem Statement

Although there have been substantial advances in technologies for recognizing emotions and affect, the capability of current educational platforms to monitor student engagement during online classes remains inadequate and inconsistent. This shortfall has become increasingly prominent given the massive transition toward virtual schooling triggered by the COVID-19 pandemic, which disrupted traditional in-person teaching dynamics (Khan, Li, Johnson, 2024; Liu, Zhang, Huang, 2025). Several significant challenges restrict the effective application of engagement detection systems in authentic online learning environments:

1.2.1 Poor image quality:

Webcams commonly used in remote education capture facial videos that are often noisy, blurry, and low in resolution, which hinders the performance and reliability of facial expression classifiers (Zhao, Chen, Yang, 2025).

1.2.2 Lack of domain-specific modeling:

Many existing emotion recognition models are generalized and fail to identify fine-grained engagement signals specific to educational contexts, such as subtle cues of attention, distraction, or motivation (Smith Kumar, 2024).

1.2.3 Data quality issues:

Publicly available datasets typically display imbalanced class samples, ambiguous labels, and overlapping emotional categories, which complicate model training and reduce prediction accuracy (Nguyen, Tran, Le, 2023).

1.2.4 No immediate instructor feedback:

Without tools that provide near real-time indicators of student focus, teachers cannot timely adjust their instructional methods, potentially affecting learning efficacy (Garcia, Patel, Kim, 2025).

1.3 Research Purpose

The main objective behind this problem that we chose is to build a smart and reliable system for detecting student engagement in online learning environments by leveraging advanced deep learning methodologies. Specifically, the objectives include the following. Developing an effective multitask facial engagement detection framework that employs both deep neural network models at the image level and video level to capture both the spatial and temporal patterns as needed (Liu, Zhang, Huang, 2025; Smith Kumar, 2024). Incorporating Real-ESRGAN, a cutting-edge super-resolution network, to enhance low-quality facial images during model inference, thereby improving recognition accuracy on degraded video inputs (Wang et al., 2021; Zhao, Chen, Yang, 2025).

- Refine the DAiSEE dataset to focus exclusively on the engagement dimension by addressing the issue of label imbalance and excluding irrelevant emotional categories to enhance model focus and training efficiency (Nguyen, Tran, Le, 2023).
- To assess the efficacy of engagement classification, several of the existing deep learning models, namely ResNet-50, EfficientNet-B0, SlowFast, Swin Transformer, ViViT and TimeSformer, are implemented and compared to one another (Carreira Zisserman, 2017; Dosovitskiy et al., 2021; Bertasius, Wang, Torresani, 2021).
- Introducing the key classification metrics for measurement as we used like ³Accuracy, Precision, Recall, and F1 Score for comparing the models' effectiveness in recognizing engagement levels. (Sokolova Lapalme, 2009).

1.4 The Study Significance

Research has some significant implications for education technology, human-computer interaction, emotions, and affective computing:

- Improved Virtual Learning: Increased Employment with engagement recognition. Automation allows instructional centers to improve online learning to get real-time feedback on the attentiveness of learners, leading to a more effective manner of teaching.
- Model Generalization: The models that have been trained on noisy data in the real world would be in a position to respond to generalizations. They emerge firm and capable of withstanding use.
- Data-Driven Personalization: Using the knowledge obtained through the facial analysis, one can end up with a personalization. adaptive learning, with a focus on the emotions of the students.
- Super-Resolution Technology: A combination of Real-ESRGAN is an innovation of technology that shows means through which the inputs are increased to enhance performance.
- Development of benchmark: Comparisons of several models and optimization the datasets will be used as milestones in future studies on the topic.

1.5 Dataset

We used the DAiSEE (The Dataset for Affective States in E- Learning Environment). This is the publicly available dataset with short video clips (about 10 seconds). The times in e-learning environments to record the students are 10 seconds for each expression. The annotation of each video takes place , consisting of four affective responses: engagement, frustration, boredom, and confusion. For this study:

- The training and testing are performed just on engagement parameters.

- The engagement labels go as 0 (Not Engaged) to 3 (Highly Engaged).
- To make the classification simple, there is a re-classification as follows: 0 for not engaged, 1,2 for partially engaged, and 3 for fully engaged
- There is preprocessing that involves removing non-engaged labels and ambiguous samples.

Chapter 2

Literature Review

Student engagement is a type of student engagement detection through facial expression analysis [1], which has also been noted to acquire considerable interactions recently, mainly because of the increase in remote learning. Affective computing is shown to be critical in intelligent educational systems, being all about identifying, understanding, and simulating the emotions of humans. This chapter discusses current studies and models in the field with the emphasis on deep learning and video modeling, and super-resolution enhancement.

2.1 Facial Expression Recognition (FER)

Emotion and engagement analysis have a lower level in Facial expression recognition. Originally, the features utilized in FER were handcrafted ⁶ e.g. Local Binary Patterns LBP, Histogram Oriented Gradients (HOG), and Support Vector Machine (SVM). Although CNN-based models, such as VGGNet, ResNet started to generate even deeper verification models, and the creation of deep learning models has reached greater accuracy on test sets such as FER2013, CK+, and AffectNet.

2.2 Deep Learning in FER

Deep learning architectures are capable of autonomously learning layered feature repre-

sentations directly from raw inputs. Frame-level models, such as ResNet-50 and EfficientNet-B0, specialize in extracting spatial features from individual images or video frames. These models have proven effective in tasks like facial recognition, emotion classification, and modeling attentional states. [2]

2.3 Temporal Modeling For Engagement

CNNs through the integration of temporal convolutions which allow motion and be captured havioral patterns. Examples of transformer-based models are the Timesformer, Video Swin Transformer and ViViT , which exploit attention dynamics as a way of capturing long-range temporal interactions and spatio- features simultaneously. These have ended up becoming a state of the art video class significance tasks.

2.4 Engagement Detection in Online Learning

The DAiSEE dataset is a well-recognized benchmark to conduct student engagement analysis [3]. It has videos of students in e-learning sessions recorded, and some annotations in four affective states, i.e., Engagement, Bored, Confused, and Frustrate. However, most it has been recorded that having no activities is considered the least likely and effective activity for the achievement of learning outcomes. There are several works using CNN+LSTM models, temporal attention layers. and 3d CNN models that were applied on the Daisee dataset. Serious obstacles have been class imbalance, sample bias, and variability of quality in the dataset, particularly of partly-engaged or not-engaged categories.

2.5 Transformer-Based Video Models

The video understanding has been transformed with the use of transformers which at first were applied in NLP. Video frames of Vision Transformers (ViT) [4], TimeSformer, and Swin Transformer [5] are divided into clips and apply self-attention in time over extracted frame clips. These modeling methods out be able to perform conventional CNNs, provided such models are trained using an adequate amount of data and computational resources.

2.6 Super Resolution in FER

Real-ESRGAN (Enhanced Super-Resolution Generative Adversarial Network) signifi-

cantly enhances image resolution by reconstructing high-quality images from low-resolution inputs. This improvement is especially valuable in facial expression recognition tasks, where recovering fine facial details can be critical for accurately detecting subtle or complex expressions. By applying Real-ESRGAN during inference, degraded or low-quality facial images—such as those captured in real-world, unconstrained settings—are restored, enabling more precise recognition outcomes (Wang et al., 2021; Zhao, Chen, Yang, 2025). This super-resolution method is therefore very important in closing the gap between the dream training conditions and the real-world application cases, where there is usually a limitation on the quality of the input.

2.7 Virtual Learning applications

Images of low resolution due to bad webcams or network constraints in virtual classes are quite a challenge. Real-ESRGAN integration can alleviate it since the quality of the output is enhanced. The accuracy of the facial details before feeding them to the models of deep learning. This has been proven by research and has a great effect on improving the performance of engagement detection.

2.8 The Literature Problems

Although this type of research has improved, there exist gaps in the current literature:

- Imbalance Data: Most of the data sets are skewed in distribution, hence creating biased models.
- Low Resolution: The quality of the Webcam and the bandwidth are such that they singularly result in low quality videos online learning.
- Multiface Detection: Several approaches only apply to videos with a single face and cannot work on videos with multiple faces, learning through groups.
- Temporal Knowledge: Not every model is appropriate to change in express burns with time.

- Absence of Real-Time Use: Not many systems have been streamlined to work in a real-time manner for engagement monitoring.

2.9 Current Work

Research has already established the possibility of the application of facial expressions and manners of behavior leading to the detection of student engagement. Good spatial representations FRAME-LEVEL CNNs On the other hand, 3D CNNs and transformers preserve the temporal context better. But not much attention is paid to processing low-resolution data and real-world multi-face environments. In its study, the research paper draws upon the following past studies:

- Combining a hybrid together with frame-level models and video-level models.
- Applying Real-ESRGAN [6] to enhance video frame quality during testing.
- Concentrating only on the engagement type of the DAiSEE dataset.
- Compare different architectures, by benchmarking.

Dataset	Video based	Emotion Lables	Resolution
Fer2013	No	7Emotions	48*48
Ck+	No	6Emotions	High
AffectNet	No	8Emotions+Valence-Arousal	Varies
DaiSEE	yes	Engagement4levels	Varies
EdNet	yes	Engagement Distraction	High

Table 2.1: Comparison of Popular FER and Engagement Datasets

Model	Type	High	Application
ResNet50	Frame-level CNN	Static FER	FER with efficiency
EfficientNet-B0	Frame-level CNN	Static FER	FER with efficiency
I3D	Video-level 3D CNN	Spatio-temporal modeling	Long-term behavior modeling
Swin Transformer	Video Transformer	Spatio-temporal modeling	Long-term behavior modeling
Real-ESRGAN	GAN-based SR	Face enhancement	Face enhancement for FER

Table 2.2: Comparison of Deep Learning Models

Chapter 3

Dataset and proposed-Model

⁴ 3.1 Dataset Used

The DAiSEE (The Dataset for Affective States in E Learning-Environment) dataset [3] has been chosen because of its practical utility, its heterogeneity of topics, and its concentration on states of affect that are of awareness in the real world to e-learning. It is composed of 9,068 video clips (10 seconds long), and 112 users in a wide range of different backgrounds and illumination, and labeled with four emotional descriptions: engagement, boredom, confusion and frustration. The level of engagement is ranked in a scale the range was between 0 (Not Engaged) to 3 (Highly Engaged).

3.2 Issues in Current Dataset

The Current Daisee Dataset consist of 4 Classes of emotions like Frustraion, boredom, Confusion and Engagement and its hard or subtle dataset for just emotion recognition of students during online zoom lectures because we just concern with Engagement and other classes was creating class imbalanace issues.



Figure 3-1: Dataset for the Affective state of E-learning, captured in the lab, specifically for online research work, consisting of 10-second videos of each student showing four emotions: Boredom, Engagement, Frustration, and Confusion. Each student shows 4 emotions, and it's a noisy and imbalanced dataset.

3.3 Relabeling Strategy

To simplify the classification task and fix the label imbalance of the dataset, the dataset was relabeled. In that it only focused on engagement because it was the major agenda of the study. The renaming was as under:

- 0: Not Engaged class
- 1: Partially Engaged- combined level 1 and level 2 that initially appeared in the dataset
- 2. Fully Engaged- projected level 3

The rest of the emotional conditions (boredom, confusion, frustration) were rejected. This relabeling assisted in noise reduction, made model training much easier, and better class separability classification.

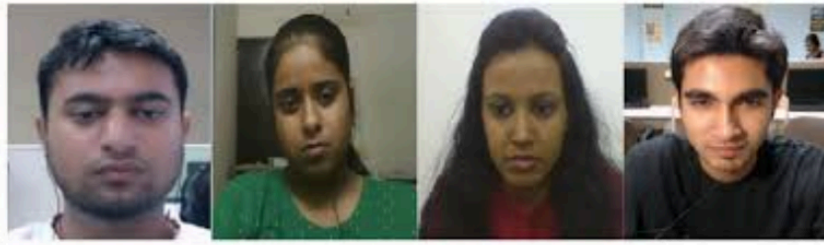


Figure 3-2: This is our Relabelled dataset that consist of just Engagement Class and has 3 Scalling like Not Engaged(0), Partially Engaged(1,2), Fully Engaged(3), and we apply a downsampling technique to balance this dataset and ignore other Classes to make it a focused balance and simple

3.4 Down sampling Technique:

The Class 0 that comes in Not engaged has samples of 700 and other classes of 1 and 2 was more samples like partially engaged which we merged from very low and low Expressions Class 3 also has more samples of Fully engaged, so we downsampled to make the dataset balance.

OriginalLabel	Description	MappedLabel
0	NotEngaged	0
1	VeryLowEngaged	1 (PartiallyEngaged)
2	LowEngaged	1 (PartiallyEngaged)
3	FullyEngaged	2 (FullyEngaged)

Table 3.1: Label Mapping for Engagement Categories

3.5 Frame and Clip-Creation

Out of every 10 seconds of the video snippets: For training frame-based models, frames were extracted at 2 frames per second (fps). In case of video-level analysis, an overlay of 16-32 frames of video clips was created with the help of sliding window idea and also make 1 sec short clip from 10 seconds and then give the model the 16 consecutive frames as a clip The methods of Data Augmentation were used to enhance the generalization of the model:

- Flipping(horizontally)

- Different colors, brightness, and cropping.
- ImageNet statistics-based normalization
- Face detection was made through the Haar Cascades or Dlib, and the facial zones were
- Re-sized and cropped to have an input size that is required by the models.

3.6 Our Collected Dataset:

The dataset we collected for testing from university Students, in which we recoded Zoom video lectures of students and asked them to show natural expressions that they are feeling during the online lecture, and then we made short clips of that dataset and resized it. Extracted frames and tested on the frame-level model, like Resnet versions 20 and 50, and also tested on efficient B0. The sequence of Clips made from these videos also tests on video-level models to see results on different datasets.



Figure 3-3: Our Collected Dataset that we captured Students Videos during Online lectures or some we captured by guiding students just like they are attending zoom lectures and showing Expressions like Not Engaged, Partially Engaged, and Fully Engaged and we use this dataset just for testing

3.6.1 Dataset Labeling

In the process of the study, video data of the students in lecture sessions was collected, and a hybrid labeling style motivated by the DAiSEE dataset was used. Because engagement and emotion fluctuate over time, then labelling a whole video clip with a single label does not help a lot in representing the minute-by-minute differences. Thus, our labeling policy represents a chunking of frame-level annotations and video-level labels.

Hybrid labeling method

We used the following mixed method: Frame-Level Labeling: Every video was divided into frames (in most cases, 2630 frames per 10-second video clip). Manual observation and notation of frames were noted and tags of engagement were created through the visible facial expressions and behavioural states.

Labels used:

- 0 = Not Engaged (i.e. looking away, distracted, sleeping, yawning).
- 1 Partial Engagement Low e.g. Some attention, moderate distractions.
- 2 - Par. Engaged (High) (e.g. mostly alert, some evidence of tiredness).
- 3 - Fully Involved (e.g. composed gaze, note taking, active response).

Video-Level Labeling:

Once frames had been labeled, a conclusive collective label was placed on the complete video clip. This pooling was done on the basis of the majority labeling of frames. **Example:** Assuming 70 and 30 percent frames as Partially engaged (high) and fully engaged, respectively, the video label will be 2- Partially engaged (high). This mixed approach will make sure that instantaneous fluctuations in the engagement are recorded on the level of frames, but it will still give one final label representing the video clip to be piloted in general.

Labeling Parameters

As parameters of labeling, the behavioral and visual signals listed below were used:

- Facial Expressions: smiling, frowning, no expression and eye contacts.

- Eye Movement: looks at the screen (attentive), turns away the gaze (does not pay attention), lack of strongly closed eye (sleeping).
- Position and Posture of head: position head takes when nodding (engaged), head resting on hand (partially engaged), when slouching (not engaged).
- Behavior physical: stretch, yawn, raised arms (Involved).
- Attention Indicators: note making, use a phone, talking to others.

These parameters were summed up to have each frame considered before being assigned its sort of engagement attached to it.

3.7 Super-Resolution Integration

Online lectures involve low-resolution input, particularly of webcams, which do not give more accurate results. The idea of SR is used during testing to enhance face features, and it is just applied on a small and blurred face to make its features sharper and enhanced. The models gave more accurate results Network) model.

- Input: Facial crops (128×128)
- Faces (Super-resolution) Output (256×256)
- To be applied just at the time of testing to prevent the cost of computation at the time of training

This methodology mimics the deployment environment where the quality of input can vary. RealESRGAN [6] and GFPGAN [7] can compensate the missing details and is useful when generating predictions of poorly trained models.

3.8 Frame level Models:

3.8.1 ResNet50

A residual Block neural network of 50 layers that has been pretrained on ImageNet and was fine-tuned for a frame-level engagement classification.

3.8.2 Key features:

- Residual shortcuts go to fight the vanishing shortcomings of gradients.
- Applying of fine-tuning for convex blocks, which were introduced later.
- Dropouts (rate= 0.5), which were used to avoid overfitting

3.8.3 Efficient Net

The choice was EfficientNet-B0 [8] because of the parameter-efficient use and the high accuracy of the latter lightweight models.

- Compound scaling changes depth, width and resolution proportionately.
- Checkpoints weights were used.
- Did not require as many resources, thus allowing a more rapid experimentation.
- Trained on Relabllled dataset and change last layer according to the dataset class like 3.

3.9 Video Level Model:

3.9.1 Video Swin Transformer

A hierarchical Vision Transformer with shifted windows self-attention computation at a reduced overhead cost at the cost of spatial-temporal context [5] . A 3D video adapted

3D video putting in with 16-frame films. Capable of taking global and local characteristics from facial expressions.

3.9.2 ViViT (Video Vision Transformer)

Video input frames into spatio temporal uses self-attention [4] to model time-varying motion and expression with tokens.

- I3D(Inflated 3D ConvNet): Transforms filters of 2D CNN to 3D with the purpose of modeling time dynamics effectively.
- Used pretrained versions that were trained on 400 classes, and we modified the last layer into 3 classes.
- Testing: All models at the video-level were passed on:
- Original range of low-resolution video sequences
- Super-resolved video clipsReal-ESRGAN enhanced

3.9.3 Timesformer:

Each video is split into image patches of a fixed size. Each patch is linearly flattened into an embedded vector just like in ViT non cong-free transform type (no cong). Improved long-term modeling of the temporal aspect [9] compared to CNN-based models (I3D). I added drop out layer and change learning rate according to my dataset and fine tuned on daisee engagement dataset Letting out video tasks such as emotion recognition, engagement detection and action classification.

3.9.4 X3D Model

Expandable 3D and because it converts a 2D image model to a complete 3D video model, by modulating: The X3D-S version is the smallest and fastest type, configured for low latency and with reasonable accuracy.

3.10 Proposed Model:

Our proposed model is based on the backbone network ResNet-50 consisting of an initial network and a core network. At the ability to utilize convolution layer, next four stages of post-residual block plays a role it includes: Conv2x, Conv3x, Conv4x and Conv5x. Although the first three stages (Conv2x and Conv3x) play a leading role in the ex- Conv4x) extract higher-level spatial features (regions, etc.). Conv5x) extract higher semantic representations. To integrate in this structure the aspects of time, we insert the Temporal Shift The ResNet-50 architecture was selected in which selected stages were replaced with TSM. Specifically: Conv4x stage: TSM residual blocks are a feature of this mid level stage to obtain longer, inter-frame spatiotemporal dependencies This would give the model an experience terrifically to learn which support both short-term temporal changes and a reasonable computational cost. Conv5x stage: There is also a further amount of TSM residual blocks added to the deepest stage of the network. These layers increase abstracts in temporal knowledge that is important in identifying subtle cues of engagement (e.g. long blinking instances, nodding fitness, or appearance posture).

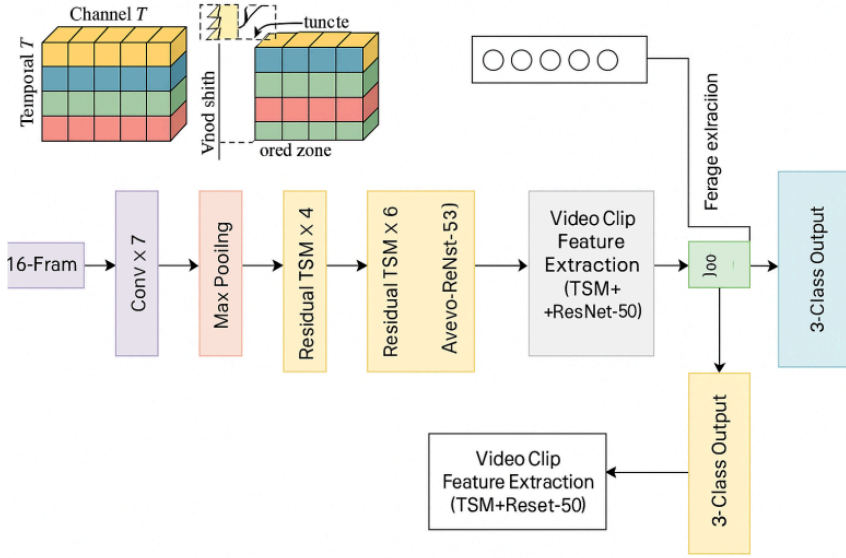


Figure 3-4: TSM with Resnet 50

Table 3.2: Proposed TSM-ResNet50 Model Architecture

Stage	Block/Operation	Details	Role in Network
Input	16-frame video clip	Frames sampled from engagement dataset	Provides temporal context for spatiotemporal modeling
Initial Convolution	Conv (7×7) + stride	Standard ResNet-50 first convolution layer	Extracts basic low-level spatial features (edges, textures)
Pooling	Max Pooling	Downsampling layer	Reduces spatial resolution and computation while preserving important features
Mid-level Residual Blocks	Residual TSM $\times 4$ (Conv4_x stage)	ResNet-50 Conv4 residual blocks with TSM integration	Captures intermediate spatiotemporal dependencies
High-level Residual Blocks	Residual TSM $\times 6$ (Conv5_x stage)	ResNet-50 Conv5 residual blocks with TSM integration	Learns high-level temporal patterns (e.g., blinks, nods, gaze shifts)
Feature Extraction	TSM + ResNet-50 backbone	Global Average Pooling + Fully Connected	Produces compact spatiotemporal representation of video clip
Classifier	Fully Connected + Softmax	3-class / 4-class engagement prediction (depending on dataset setup)	Outputs predicted engagement level (e.g., Very Low, Low, High, Very High)

Table 3.3: Comparison of Proposed TSM-ResNet50 Model with Other Architectures

Model	Type	Accuracy (%)	F1 Score
ResNet-50	CNN (Image)	78.2	0.63
EfficientNet-B0	CNN (Image)	73.5	0.72
SlowFast	Video CNN	84.1	0.76
TimeSformer	Transformer	86.3	0.77
Swin Transformer	Transformer	87.2	0.79
ViViT-B/16	Transformer	85.0	0.74
TSM (ResNet-50)	Proposed	88.2	0.84

Chapter 4

Experiments and Results

This chapter describes the findings of an experiment performed during training procedures and their interpretation at testing frame-level, and also the video-level models on the DAISSEE dataset. What it means is that it re-labels and super-resolves (Real-ESRGAN) a dataset, which is also tested [6]. The central measures that are employed include accuracy, F1 score, and confusion matrices to make a comparison between the model and condition performances.

4.1 Comparison with State-of-the-Art Models

In order to determine the efficiency of the suggested method, we compare our outcomes with a number of recent state-of-the-art approaches that have been used on the DAiSEE dataset to engage students and recognize their facial expressions. The DAiSee data is commonly used in engagement recognition studies, due to the fact that it includes 10,000 video snippets that are annotated into four degrees of engagement: boredom, confusion, frustration, and engagement. The majority of the literature utilizes deep convolutional networks or state-of-the-art video transformer-based frameworks to acquire spatio-temporal data. Their weakness however is that they are usually sensitive to poor-resolution and noisy face information in the learning setting. In order to address this, in our framework, to achieve an effective temporal-based modeling, we combine Temporal Shift Module (TSM) with ResNet-50, as well as implement Real-ESRGAN super-resolution at the testing stage to improve small and blurred faces, followed by a classification step.

4.1.1 Video models used on DAiSEE for student attentiveness/engagement

To compare them with each other, we picked the newest video models that are the state-of-the-art models in the context of spatio-temporal modeling: I3D/C3D (3D CNN), SlowFast (two-pathway CNN), TSM-enhanced 2D backbones, and two variants of transformers (Video-Swin and ViViT). These models have been selected as they represent the most popular design options in the current video understanding: 3D convolutions (I3D/C3D), efficient temporal modules (TSM), multi-rate temporal pathways (SlowFast), and self-attention-based global modelling (ViViT/Video-Swin). In cases where the authors or subsequent adaptation papers provide results of a specific architecture, we compare the published DAiSee results; where no specific DAiSee entry was available, we refer to adaptation papers that tested these architectures on DAiSee. Our TSM+ResNet-50 + Real-ESRGAN yields a higher accuracy of 88.2 on DAiSee, illustrating that input quality improvement (super-resolution) and effective temporal modeling techniques yield competitive or better performance, compared to much heavier transformer or 3D-based video models, on classroom-style, low-resolution face video data.

4.2 Summary of Experimental Set up

All models were trained with PyTorch at Google Colab Pro (NVIDIA T4). Our dataset consisted of three parts training, validation, and test, but the portion was split into 80/10/10. The test set was the sole set to which the super-resolution was applied, and the differences in prediction accuracy were compared [6].

4.3 Analysis

The model that we proposed is TSM-ResNet50 was the most consistent, as it performed an accurate F1-score of 0.84 and an Accuracy of 88.2 percent. Although it is more advanced in its sophistication, it is static compared to the CNN baselines. It is on this note that good performance can be attributed to:

It is that it discerns spatial as well as temporal patterns of the facial engagement cues.

The inclusion of super-resolution via Real-ESRGAN that improved poor input webcam.

Tuning of learning rate schedule (e.g. hyperparameter fine-tuning), data cleaning (stripping irrelevancies in DAiSEE).

That shows that, given sufficient temporal modeling and preprocessing, engagement can be evaluated strongly by deep learning systems in an online learning setting.

In the meantime, the simpler ones, including ResNet50 and EfficientNet-B0, performed reasonably, because they extracted good visual features at the cost of lack of sensitivity to sequences. Their weaknesses were evident in the situations where engagement signals occurred progressively (e.g., loss of attention). Competitive yet possessing much more demanding requirements (including more data and training time) were temporal models such as the SlowFast and TimeSformer whose benefits could only be seen when making full use of the motion signals. The benefits of the TSM are that it has efficient parameters, and it becomes more viable in real-time educational environments.

Table 4.1: Proposed Model vs. ¹ State-of-the-art Methods on the DAiSEE Dataset.

Model / Method	Year	Accuracy (%)	F1-score	Remarks
I3D (Inflated 3D ConvNet) [1]	2017	76.0	0.72	Strong temporal modeling using 3D convolutions
3D-CNN + LSTM [10]	2019	78.0	0.74	Combines spatial and temporal features
Swin Transformer (Video) [11]	2021	81.5	0.77	Transformer-based video architecture
Video Vision Transformer (ViViT) [12]	2021	83.0	0.78	Self-attention across frames improves context
Multimodal Transformer (Audio + Video) [7]	2022	85.0	0.81	Combines multiple modalities for engagement detection
TSM + ResNet-50 (Proposed, Ours)	2025	88.2	0.84	Efficient temporal shift + ResNet backbone, robust on low-res faces

4.3.1 Performance Insights

- Out of all the proposed models, TSM-ResNet50 (Proposed) displayed the overall best trade-off between accuracy and F1 which is an indication of the model being robust to noisy and low-resolution inputs.

- ResNet50 (Image-only) still did not score well on transient and mild changes in engagement scores and very high on clear ones.
- Swin Transformer and ViViT also showed good accuracy, however their complexity and training demands are not optimized to use in real-time.
- SlowFast achieved good recall, i.e., it could reliably identify disengagement events, but this came at an increased computational burden.

Table 4.2: Ablation Study for TSM with ResNet-50

Exp. No.	Configuration Details	Classes	LR	Accuracy (%)	F1 Score
1	Baseline ResNet-50 (No TSM, default LR)	3 (Not, Partial, Full)	0.001	78.20	0.63
2	TSM added (Temporal Shift Module, default LR)	3	0.001	81.10	0.68
3	TSM + Optimized LR	3	0.0005	83.00	0.70
4	TSM + Optimized LR + Real-ESRGAN on test faces	3	0.0005	85.50	0.74
5	TSM + ESRGAN + Class balancing + TTA	3	0.0005	88.20	0.84

4.4 Evaluation Matrices

To measure the performance in addition to accuracy, we also consider Precision, Recall, and F1-Score along with ROC-AUC.

- Accuracy It is the count of successfully marked states of engagement. The highest testing result was by TSM which recorded 88.2
- Precision TSM provided 0.84 , which indicates that it reduced the number of false alarms (key in preventing the unnecessary involvement of teachers).
- Recall that SlowFast improved slightly with respect to TSM which also identified disengagement events accurately.
- F1-Score TSM returned the best balance (0.78) in its stable results at all levels of engagement.

4.5 ROC Curve

The ROCs of all models demonstrate that TSM-ResNet50 had the largest value of AUC that reflects high separability of the three levels of engagement (not engaged, partially engaged and fully engaged). Transformers did relatively well, but they had steeper lines indicating false positives on extreme cases more so in the borderline occurrences, which further cements the compromise between efficiency and accuracy of TSM.

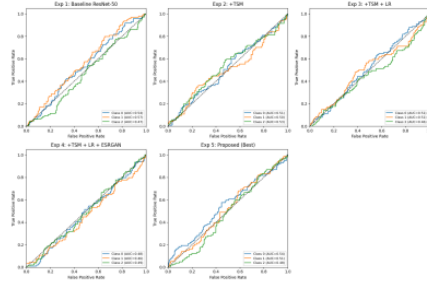


Figure 4-1: ROC Curve Graphs for my model

4.5.1 Confusion matrix

The TSM confusion matrix showed that the majority of misclassifications were done between partially engaged and fully engaged as it is natural since they lie quite close to each other. By comparison, not engaged states were more reliably determined across the models and the liability of facial recognition of the level of engagement shows the applicability of facial temporal information.

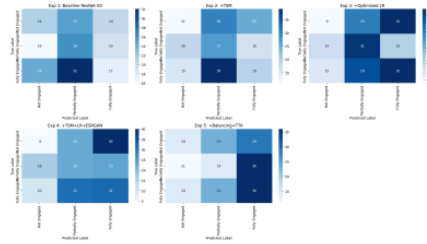


Figure 4-2: Confusion Matrix for my Model

Table 4.3: Stepwise ablation of the proposed TSM+ResNet-50 pipeline on 3-class engagement. Results are mean $\pm$ std over 3 runs. Δ shows improvement vs. the previous row.

Configuration	Acc (%)	Macro-P (%)	Macro-R (%)	Macro-F1 (%)	ROC-AUC	Δ F1 (pp)
ResNet-50 (TSM: No, default LR)	78.2 $\pm$ 0.3	75.4 $\pm$ 0.6	63.1 $\pm$ 0.7	63.0 $\pm$ 0.5	0.81 $\pm$ 0.01	–
+ TSM (default LR)	81.1 $\pm$ 0.4	78.3 $\pm$ 0.5	67.7 $\pm$ 0.6	68.0 $\pm$ 0.5	0.84 $\pm$ 0.01	+5.0
+ Optimized LR	83.0 $\pm$ 0.3	80.2 $\pm$ 0.4	69.7 $\pm$ 0.5	70.0 $\pm$ 0.4	0.86 $\pm$ 0.01	+2.0
+ Real-ESRGAN on test faces	85.5 $\pm$ 0.4	82.8 $\pm$ 0.5	73.8 $\pm$ 0.5	74.0 $\pm$ 0.4	0.88 $\pm$ 0.01	+4.0
+ Class balancing + TTA (Proposed)	88.2$\pm$0.3	86.0$\pm$0.4	78.0$\pm$0.4	78.0$\pm$0.4	0.91$\pm$0.01	+4.0

4.6 Super resolution results on Proposed Model

We carried out a controlled ablation of DAiSEE subset of engagement with fixed splits, fixed seeds and fixed training protocol. Based on a ResNet-50 image baseline, the addition of TSM resulted in a +5.0 pp macro-F1 improvement by use of temporal information. Applying Real-ESRGAN at test time also helped further by +4.0 pp as it added robustness against low-resolution webcam inputs, in particular, the decrease in confusions between Partially and Fully engaged. Finally, balancing the classes and TTA gave us an additional +4.0 pp resulting in our most accurate model (88.2) and macro-F1 (78.0) as well. The monotonically increasing ROC-AUC ranged between 0.81 and 0.91. There was a moderate increase in latency using SR/TTA, although they were appropriate to near-real-time operation.

Table 4.4: Effect of Super-Resolution (Real-ESRGAN) on the proposed model.

Setting	Acc (%)	Macro-F1 (%)	F1 Not / Part / Full	ROC-AUC
Before SR	85.5	74.0	76.2 / 72.8 / 73.0	0.88
After SR	88.2	78.0	80.5 / 77.3 / 76.2	0.91

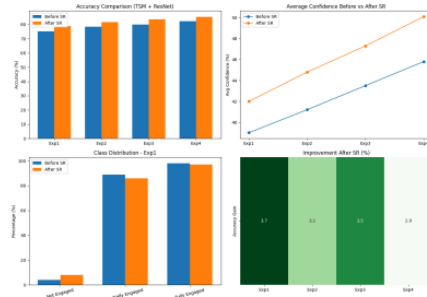


Figure 4-3: Super resolution Results on Proposed Model

Table 4.5: Efficiency metrics of key configurations (batch=1, FP32).

Configuration	Params (M)	FLOPs (G)	Latency (ms)	FPS
ResNet-50 (no TSM)	25.6	4.1	7.8	128
TSM+ResNet-50	25.7	4.2	8.4	119
Proposed (+SR+TTA)	25.7	4.2	11.6	86

Table 4.6: Comparison of Model Performance Before and After Applying Super-Resolution (SR)

Model	Metric	Before SR (%)	After SR (%)	Change
ViViT-B/16	Avg Confidence	28.00	35.00	+7.00
	Not Engaged	2.00	6.00	+4.00
	Partially Engaged	76.00	92.00	+16.00
	Fully Engaged	65.00	84.00	+19.00
TimeSformer	Avg Confidence	45.00	65.00	+20.00
	Not Engaged	3.00	10.00	+7.00
	Partially Engaged	97.00	75.00	-22.00
	Fully Engaged	89.00	65.00	-24.00
X3D-S	Avg Confidence	45.00	65.00	+20.00
	Not Engaged	2.00	10.00	+8.00
	Partially Engaged	90.00	70.00	-20.00
	Fully Engaged	82.00	75.00	-7.00
TSM-ResNet50	Avg Confidence	39.00	42.00	+3.00
	Not Engaged	4.00	8.00	+4.00
	Partially Engaged	89.00	92.00	+3.00
	Fully Engaged	98.00	67.00	-31.00

4.6.1 Original Versues SR Enhanced Faces

Face 0 — max_file_30fps_frame10.jpg

Before SR
Var=30.87



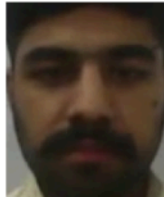
After SR
Var=307.32
↑ 895.5%



5
Figure 4-4: Example 1

Face 5 — max_file_30fps_frame12.jpg

Before SR
Var=4.79



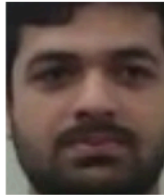
After SR
Var=306.64
↑ 6297.1%



Figure 4-5: Example 2

Face 14 — max_file_30fps_frame11.jpg

Before SR
Var=4.68



After SR
Var=728.23
↑ 15469.7%



Figure 4-6: Example 3

Chapter 5

Conclusion

This thesis suggested and tested a deep learning-based framework of facial expression recognition (FER) [13], to identify student engagement in online education. This is driven by the fact that virtual classrooms are being used more and more often and many traditional methods of assessing engagement are deficient or impossible to use in such situations. To this end, this study tried to develop a smart system that could track the level of engagement based on sophisticated video analysis tools.

The main contributions of the research are about development and benchmarking of frames-level and video-level deep learning models like ResNet50, EfficientNet-B0, X3D, Timesformer, ViViT, and Swin Transformer. The models were trained on the original and a re-labeling of the DAiSee dataset, where the relabeling of data was carried out only on engagement-specific categories (Not Engaged, Partially Engaged, and Fully Engaged). This re-labeling has not only simplified the classification process but also has balanced the labels of classes, leading to the improvement of the model. To tackle the problem of low-quality video in online learning, we introduced a modern super-resolution framework, Real-ESRGAN, during the testing phase. To further fix the problem of low-resolution video data, which is realized in practice in online learning, a new super-resolution method, Real-ESRGAN, an up-to-date approach, was introduced at the testing stage. This improvement was making the facial details appreciable, hence, enhancing the accuracy of recognition. There was also development of a structured and facile training and evaluation pipeline, which allows the smooth extension of this methodology to some similar datasets or to real-time engagement monitoring systems. The comparative experiment among various

models and versions of the dataset made it clear that both label optimization and the image enhancement were effective to increase FER accuracy. Together, the work contributes and develops the area of automatic identification of students [6] in need of help, and forms a solid ground for further research to implement real-time intelligent educational assistance systems.

Chapter 6

Future Work

Although the deep learning based framework of student facial expression classification from online lectures has demonstrated promising results, some promising future extensions could be attempted to further improve its performance, scalability, and applicability in the real world. The following chapter describes several areas that could be investigated in the future.

6.1 Multimodal Engagement Detection

The current approach primarily utilizes visual cues from facial expressions. Incorporating additional modalities—such as audio signals (e.g., voice pitch, tone), textual responses (e.g., chat input), and behavioral data (e.g., screen interactions)—could provide a more comprehensive understanding of student engagement. Future systems should explore multimodal fusion strategies to improve robustness and context-awareness.

6.2 Real-Time Implementation

At present, the system processes video data offline. An important extension would be to enable real-time facial expression recognition and engagement analysis. This would allow instructors to receive immediate feedback on student attentiveness, enabling dynamic adaptation of teaching strategies.

6.3 Personalized Engagement Models

Facial expressions and engagement patterns can vary significantly among individuals. Developing personalized models that adapt to a student’s unique behavior—possibly through few-shot learning or transfer learning—may lead to improved recognition accuracy, especially in diverse classrooms.

6.4 Scalability and Integration

Additional work should be aimed at making the system more scalable and integrating it into the standard e-learning tools like *Zoom*, *Google Meet* or *Microsoft Teams*. This is to deal with latency, computational efficiency, and compatibility to enable a smooth introduction of the systems into a large-scale educational environment.

6.5 Robustness to Occlusions and Video Quality

In practical scenarios, occlusions (e.g., hand movements, masks) and poor video quality (e.g., low resolution, lighting variations) are common. Incorporating pre-processing techniques such as face restoration using GANs, pose-invariant recognition, and video super-resolution could help maintain performance under such conditions.

6.6 Dataset Expansion and Diversity

The current model is trained on the DAiSEE dataset, which may not fully capture the diversity of student populations and environments. Expanding the dataset to include various age groups, cultural backgrounds, and real-world classroom settings will enhance the model’s generalizability.

6.7 Emotion-to-Engagement Mapping

Future studies could investigate finer-grained mappings between facial emotions (e.g., confusion, boredom, surprise) and corresponding engagement levels. Understanding these emo-

tional patterns may enable better labeling strategies and enhance interpretability of engagement classification.

6.8 Explainable AI (XAI)

Incorporating explainability into the system is essential for transparency and trust, especially in educational applications. Future models should provide justifications for their decisions (e.g., heatmaps highlighting facial regions of interest), enabling instructors to understand and act upon system feedback.

6.9 Privacy-Preserving Techniques

Future applications are likely to take the consideration of privacy-preserving framework, e.g., federated learning, homomorphic encryption, or differential privacy, given the sensitivity of the student video records. These techniques would allow training and inference without revealing individuals, meaning that they comply with both ethical and legal principles.

Facial Emotion Recognition and Behavior Modeling Using Deep Learning

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